

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

BIOLOGY

8875/02

Paper 2 Core Paper

1 September 2014

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

Circle the question number of the question attempted

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
5 / 6	
Total	

This document consists of **12** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Phospholipase A2 (PLA2) is an enzyme that hydrolyses the bond between glycerol and fatty acid found in phospholipid. Venom from both snakes and insects is a stimulant of PLA2. Due to the increased activity of PLA2 resulting from a snake or insect bite, arachidonic acid (a type of fatty acid) is released from the phospholipid membrane disproportionately.

Fig. 1.1 shows the final three dimensional conformation of PLA2.

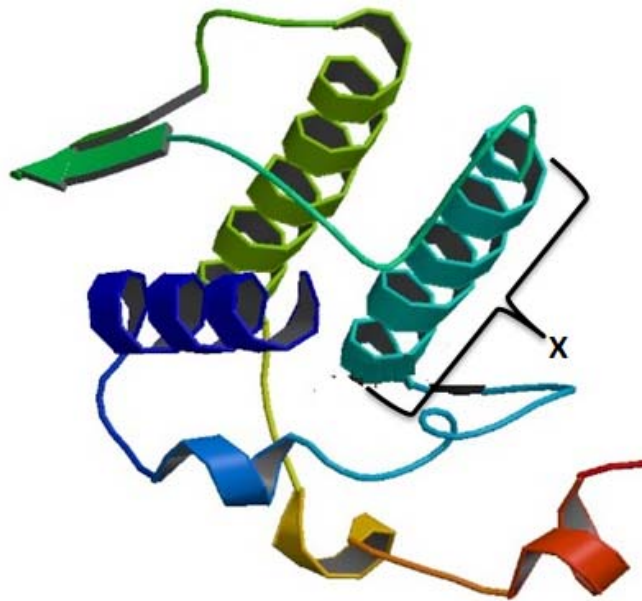


Fig. 1.1

- (a)** State the bond that is hydrolysed by PLA2. [1]

- (b)** Name and describe the secondary structure in the region labelled X. [3]

(c) Describe how PLA2 assumes its final conformation. [4]

- (d) Bromoenol lactone is known to inhibit PLA2. The effect of bromoenol lactone on the activity of PLA2 is shown in Fig. 1.2

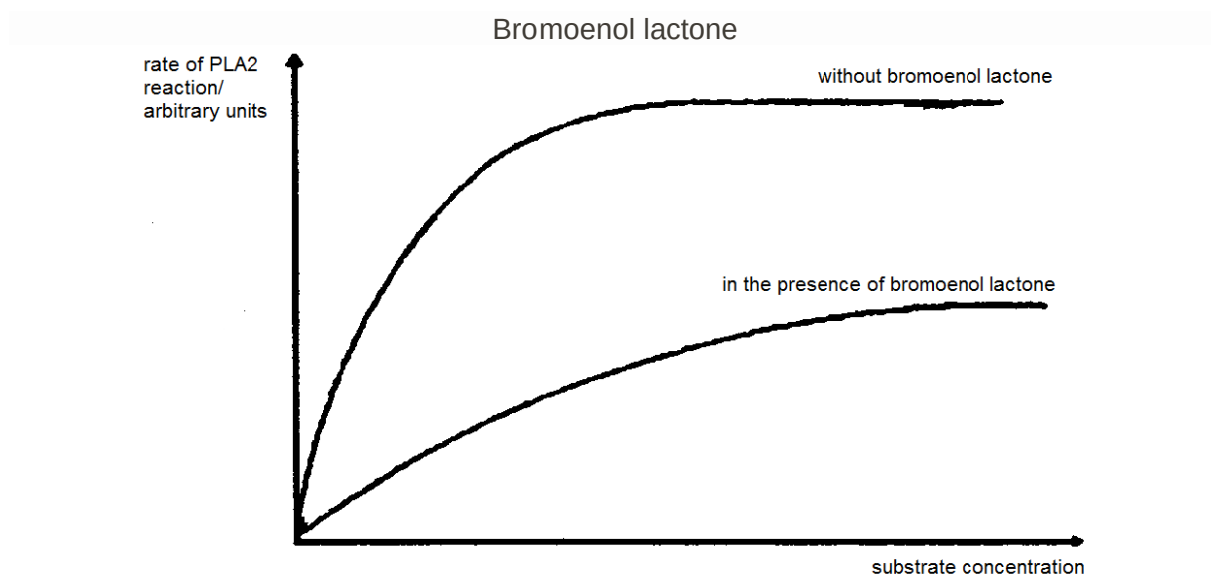
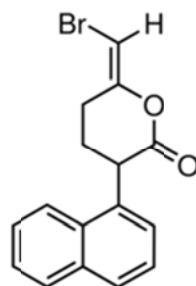


Fig. 1.2

With reference to Fig.1.2, explain the effect of bromoenol lactone on PLA2. [4]

[Total: 12]

- 2 DNA contains the genetic information needed for protein synthesis. Fig. 2.1 shows a stage in translation.

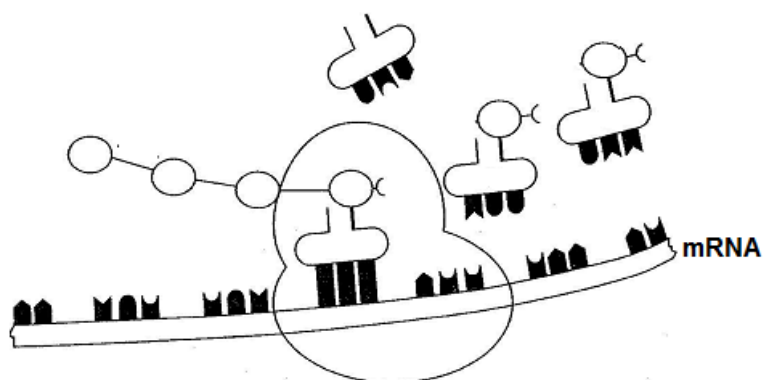


Fig. 2.1

- (a)** Explain how tRNA binds to mRNA. [2]

- (b)** Describe how the process shown in Fig. 2.1 is terminated. [4]

Fig 2.2 shows how DNA extracted from two individuals can be analysed.

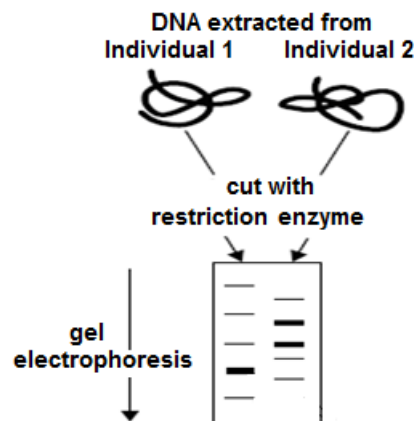


Fig 2.2

(c) Explain why the DNA of each individual produces a different band pattern after gel electrophoresis, even when the same restriction enzyme was used. [3]

(d) State how DNA can be visualised after separation using gel electrophoresis. [1]

[Total: 10]

3 Fig. 3.1 shows an electron micrograph of a chloroplast.

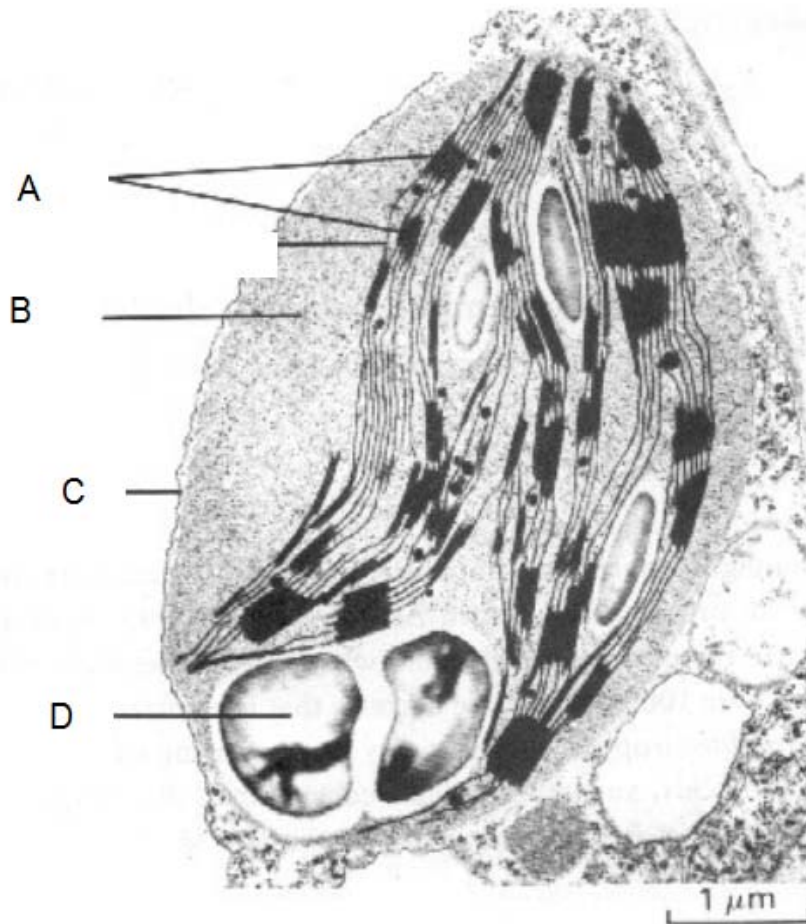


Fig.3.1

(a) Identify structures/regions A to D. [2]

- A _____
- B _____
- C _____
- D _____

(b) ATP synthase is an enzyme that synthesizes ATP from ADP and inorganic phosphate (P_i). Fig. 3.2 shows the structure of ATP synthase. It is found in both the chloroplast and the mitochondrion.

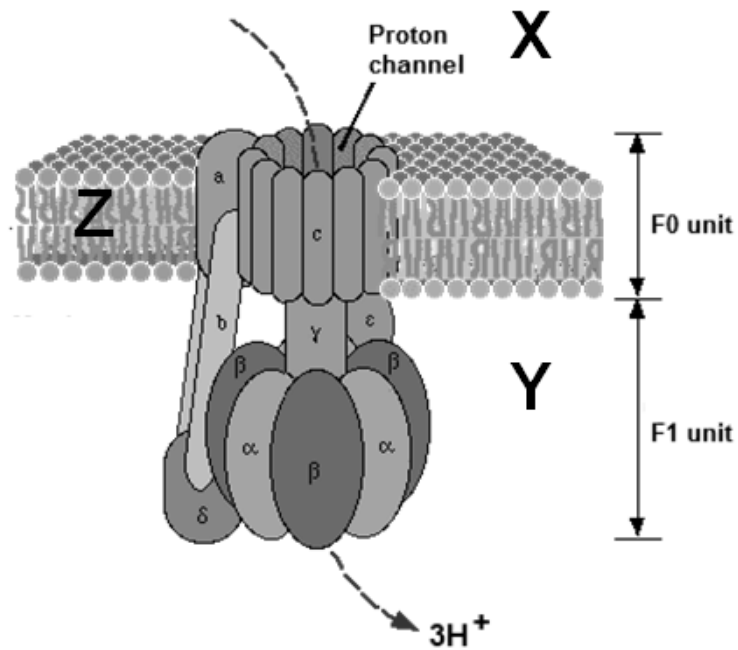


Fig.3.2

Identify the regions (X, Y, Z) as found in the mitochondrion and chloroplast respectively. [2]

	X	Y	Z
Mitochondrion			
Chloroplast			

In both organelles, a proton gradient exists across the membrane where ATP synthase is located. The synthesis of ATP by ATP synthase is coupled with the diffusion of H^+ across the proton channel down its concentration gradient.

(c) Describe how this proton gradient is generated in the chloroplast. [3]

(d) Paraquat is a weedkiller. It binds with electrons of PS I.

Suggest how paraquat results in the death of a plant. [2]

[Total: 9]

4 Fig 4.1 shows the evolution of resistant bacteria from a bacteria colony.

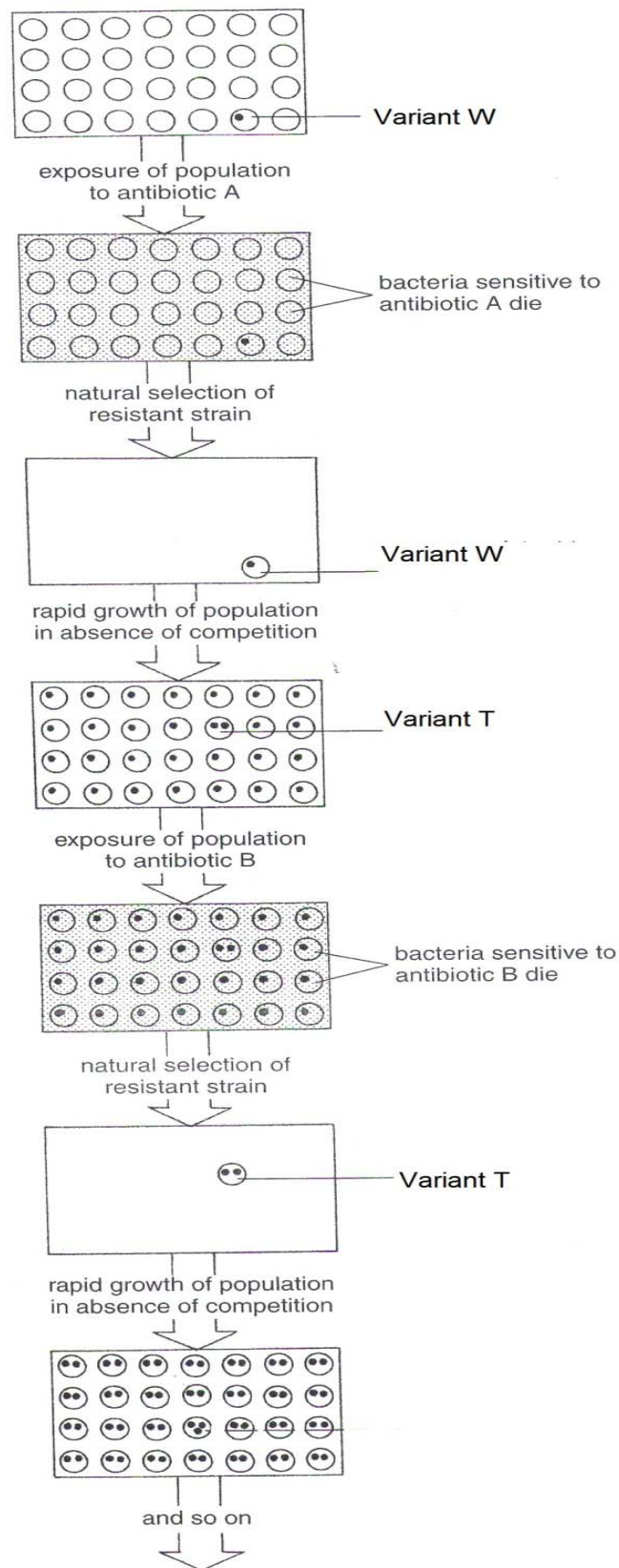


Fig. 4.1

(a) State one process which gives rise to variant W and T. [1]

(b) Explain how natural selection has led to the evolution of antibiotic resistant strains of bacteria [5]

(c) Using Darwin's theory of Natural Selection, explain the consequences of not having genetic variation in a population of bacteria. [1]

(d) Explain how DNA sequences can be used to determine evolutionary relatedness between species. [2]

[Total: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a) Outline the functions of membrane-bound organelles involved in protein synthesis and secretion. [6]
- (b) Outline the behaviour of chromosomes during mitosis. [6]
- (c) Describe the steps in polymerase chain reaction and contrast the process with DNA replication. [8]
- 6 (a) Describe, with named examples, how genetic engineering has helped in improving the quality and yield of crop plants in solving the demand for food in the world. [8]
- (b) Discuss the ethical and social implications of genetically modified crop plants. [6]
- (c) The implications for individuals and society possessing the detailed genetic information made possible by the Human Genome Project is paramount.

Discuss some ethical concerns that have arisen about the Human Genome Project.[6]